Notes on Self Avoiding Walks

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1. Introduction to SAWs

In the physics of linear polymers, it is known that in an equilibrium situation the configurational probability is given by Boltzmann's law:

$$\mathbb{P}\left(r(s), s \in (0, N)\right) = \frac{e^{-}\beta H(r(s))}{Z}\mathcal{D}r(s). \tag{1.1}$$

Where the differential $\mathcal{D}r(s)$ is understood in the path-integral sense with respect to a position variable r indexed by the length s, and the probability $\mathbb{P}(r(s), s \in (0, N))$ corresponds to the probability of a specific path. In this framework, defining a model of a polymer means defining a specific energy $-\beta H(r(s))$. The first usable Hamiltonian is the Edwards Hamiltonian, which contains only the connectivity term:

$$-\beta H(r(s)) = -\frac{d}{2b^2}\int_0^N \mathrm{d}s \left|\frac{dr}{ds}\right|^2. \tag{1.2}$$

It is interesting to observe that the same Hamiltonian is found by considering the classical random walk model in d dimensions. For this model we know how to define certain configurational exponents, among which the Flory exponent, which defines the growth of the end-to-end distance:

$$\langle R^2\rangle^{1/2} \sim N^{\nu}. \tag{1.3}$$

Which in the case of the ideal RW is $\nu = 1/2 \quad \forall d$. To this one adds the entropic exponent γ and the connectivity constant μ, which govern the number of configurations:

$$C(N) \sim \mu^N N^{\gamma-1}. \tag{1.4}$$

In the case of the ideal RW $\mu = z$ (coordination number) and $\gamma = 1$. While this model is particularly convenient to use because of its analytic solvability, several results differ greatly from experimental values, since they neglect various important properties of polymer physics. Among these, one of the most relevant is the non-self-intersection condition, which prevents two monomers from overlapping in space. To introduce this correction it is necessary to add new contributions to the Hamiltonian, which introduce a penalty for very short-range interactions:

$$-\beta H(r(s)) = -\frac{d}{2b^2}\int_0^N \mathrm{d}s \left|\frac{dr}{ds}\right|^2 - \frac{v}{2}\int_0^N \mathrm{d}s \int_0^N \mathrm{d}s' V_{exc}(r(s) - r(s')). \tag{1.5}$$

Note that this interaction is non-local, since the potential is evaluated at two distinct points of the polymer. From the point of view of the statistical random-walk process, this corresponds to the loss of the Markov property, since for each increment one has to consider the whole history of the walk. For simplicity one assumes that the potential has the form of a Dirac delta:

$$V_{exc}(r(s) - r(s')) = \delta(r(s) - r(s')). \tag{1.6}$$

A convenient way to write this new term is obtained by introducing the concentration:

$$c(r) = \int_0^N \mathrm{d}s\delta(r - r(s)). \tag{1.7}$$

In this way one obtains:

$$\int_0^N \mathrm{d}s \int_0^N \mathrm{d}s' V_{exc}(r(s)-r(s')) = \int \mathrm{d}r \int \mathrm{d}r' \int_0^N \mathrm{d}s \int_0^N \mathrm{d}s' \delta(r-r(s)) V_{exc}(r-r') \delta(r'-r(s')) \tag{1.8}$$

$$= \int \mathrm{d}r \int \mathrm{d}r' c(r) V_{exc}(r-r') c(r') = \int \mathrm{d}r \; c(r)^2. \tag{1.9}$$

In the next sections we will study some methods to estimate the new exponents of this model.

2. SAW as a stochastic process and an equilibrium system

In the previous section we saw a model that defines a configurational probability through a Boltzmann weight, which disfavours contact terms. It is worth noting that this is only a "penalty", and intersecting walks are still allowed. For this reason the model is called the *weakly self avoiding walk* (WSAW). If one wants to consider the *self avoiding walk* proper (SAW), one has to define the statistical process that assigns the same probability to all walks of fixed length, on a discrete lattice. Once this is done, if one manages to take a continuum limit of the process, one obtains a stochastic process as the length varies. As in the case of the Boltzmann weight, for the stochastic process too it is possible to associate the exponents ν, γ and μ and to talk about universality. In the case of lattice models, saying that an exponent defines a certain universality class means that it does not depend on the microscopic details of the model from which it is obtained, for example on the type of lattice with which the scaling limit is taken and on the details of the interaction. In this context one can find that μ depends on the specific model used, while ν and γ are universal exponents. In this regard, it is often possible to verify that the SAW and WSAW models belong to the same universality class (for this reason they will subsequently be used interchangeably).

3. Estimate of the Flory exponent

A first estimate of the exponent ν can be obtained by requiring that the size of the two energy terms be comparable with each other for large N. Considering:

$$-\beta H(r(s)) = -\frac{d}{2b^2} \int_0^N \mathrm{d}s \left|\frac{dr}{ds}\right|^2 - \frac{v}{2} \int_0^N \mathrm{d}s \int_0^N \mathrm{d}s' \delta(r(s)-r(s')). \tag{3.1}$$

One observes that:

- the connectivity term grows as $\sim N^{2\nu-1}$,
- the exclusion term grows as $\sim N^{2-d\nu}$;
- equating the two terms one obtains:

$$\nu = \frac{3}{2+d}. \tag{3.2}$$

Let us see case by case what this expression corresponds to:

1. In d=1 one obtains $R \sim N$. This result is exact, since once the first step is chosen the random walk is deterministic, constrained to always move forwards or backwards;

2. In d=2 one obtains $\nu = 3/4$. This result too turns out to be exact, through conformal invariance arguments (further details are discussed in section 6.1);

3. In d=3 one finds $\nu = 3/5 = 0.6$. Numerically one finds that the exact value is 0.588974.... In this section and in section 5 further arguments will be given showing the reasonableness of this estimate.

4. In d=4 one finds $\nu = 1/2$, as obtained in the case of the ideal RW. In particular in this case too the exponent turns out to be exact, since for $d \geq 4$ the self-interaction term is negligible with respect to the connectivity term and on average the chains behave ideally.

Let us now give a mean-field type argument that better justifies the result $\nu = 3/5$ of the 3d SAW. Consider the thermodynamic expression of the free energy:

$$F = U - TS. \tag{3.3}$$

As entropy, we say that to a first approximation we can associate the entropy of the ideal chain, given by :

$$S(R, N) \approx -\frac{3k_B}{2}\frac{|R_e|^2}{Nb^2}. \tag{3.4}$$

As energy, instead, we consider only the excluded-volume term in the form (1.8) and take as concentration a mean value $c(r) \approx \frac{N}{R_e^3}$. One therefore obtains:

$$U(R, N) = \frac{vk_BTN^2}{2R^3}. \tag{3.5}$$

The free energy as a function of the end-to-end distance is written:

$$F(R, N) \approx k_BT\Big(\frac{vN^2}{2R^3} + \frac{3R^2}{2Nb^2}\Big). \tag{3.6}$$

Requiring that the free energy take its minimum value as R varies (i.e. imposing $\frac{\partial F}{\partial R} = 0$) one obtains:

$$|R|^5 \sim vb^2N^3 \rightarrow R \sim N^{3/5} \rightarrow \nu = 3/5. \tag{3.7}$$

In agreement with the dimensional estimate made previously. We conclude by observing that:

- The entropy estimate is an overestimate, since in counting the configurations it also includes self-intersecting paths;

- The energy estimate is instead an underestimate, since it does not consider the connectivity term and the correlations.

4. Mean-field theory

A method to try to obtain quantitative corrections with respect to the ideal chain is to write the excluded-volume term through an external potential that simulates the effect of a mean field.

$$k_B T V_{exc} = vc(r) \tag{4.1}$$

In this case the problem turns into finding a concentration function that simulates the exclusion effect. A first method that can be adopted is iterative. In particular, knowing that the concentration can be written as:

$$c(r) = \frac{\int \mathrm{d}r' \int \mathrm{d}r'' \int dn G(r, r', n) G(r'', r, N-n)}{b^3 \int \mathrm{d}r' \int \mathrm{d}r'' G(r', r'', N)}. \tag{4.2}$$

And that on the other hand the propagator satisfies:

$$\Big(\frac{\partial}{\partial s} - \frac{b^2}{2d}\nabla_r^2 + vc(r)\Big) G(r, 0, N, 0) = \delta(r)\delta(N). \tag{4.3}$$

Which can be rewritten as:

$$G(r, r', N, 0) = b^3 \sum_k \psi_k(r) \psi_k^*(r') e^{-N e_k} \tag{4.4}$$

Where $\psi_k(r)$ and ϵ_k are the eigenfunctions and the eigenvalues:

$$\Big(-\frac{b^2}{2d}\nabla_r^2 + vc(r)\Big)\psi_k(r) = \epsilon_k \psi_k(r) \tag{4.5}$$

At this point one can start from a function $c_0(r)$, derive the corresponding propagator G_0 through (4.3), compute a new concentration $c_1(r)$ as a function of the propagator (4.2) and continue with the chain:

$$c_0 \to G_0 \to c_1 \to G_1 \to c_2 \to ... \to c_{SCF} \tag{4.6}$$

In the hope that the sequence converges to some function $c_{SCF}(r)$. Another method that uses the mean field consists in writing an implicit expression for $c(r)$ in which the propagators contain the same $c(r)$ used to define it. To obtain an explicit differential equation it is necessary to introduce an approximation for the propagator; for example one can use the ground state dominance approximation, such that:

$$G(r, r', N, 0) \approx b^3 \psi_0(r) \psi_0^*(r') e^{-N\epsilon_0} \tag{4.7}$$

From the definition, the concentration can be written as:

$$c(r) = N|\psi_0(r)|^2 =: \phi^2(r) \tag{4.8}$$

At this point we insert this concentration as a potential in the eigenvalue problem:

$$\Big(-\frac{b^2}{2d}\nabla_r^2 + v\phi^2(r)\Big)\phi(r, r', N, 0) = \epsilon_0 \phi(r, r', N, 0) \tag{4.9}$$

Obtaining an equation that is non-linear in ϕ.
Note that this equation is obtained from the variational principle of an energy with harmonic potential + ϕ^4 (double well potential):

$$F[\phi] = \int \mathrm{d}r \frac{b^2}{6}|\boldsymbol{\nabla}\phi|^2 + \frac{v}{2}|\phi|^4 - \epsilon_0|\phi|^2 \tag{4.10}$$

5. Equivalence with the O(n) model for $n \to 0$

In this section we show an interesting result, which establishes a duality between the O(n) model and SAWs. In particular we want to prove that the propagator of the self-avoiding model defined above:

$$G(r,0,N,0) = \int_{r(0)=0}^{r(N)=r} \mathcal{D}r(s) exp\Big(- \frac{d}{2b^2} \int_0^N \mathrm{d}s \left|\frac{dr}{ds}\right|^2 \Big) exp\Big(- \frac{v}{2} \int_0^N \mathrm{d}s \int_0^N \mathrm{d}s' \delta(r(s) - r(s'))\Big). \tag{5.1}$$

Laplace-transformed:

$$G(r,r',m^2,s') = \int_0^\infty \mathrm{d}s e^{-m^2 s} G(r,r',m^2,s'). \tag{5.2}$$

Corresponds to the Ginzburg-Landau-Wilson functional for the O(n) model:

$$G(r,r',m^2,s') = \int \mathcal{D}^n \varphi(x) \varphi_1(x') \varphi_1(x'') exp(-\beta H_{GLW}(\varphi)) \tag{5.3}$$

With

$$\beta H_{GLW}(\varphi) = \frac{1}{2} \int \mathrm{d}x \Big(\sum_{a=1}^n [|\boldsymbol{\nabla} \varphi_a|^2 + m^2 \varphi_a^2] + v \sum_{a,b} \varphi_a^2 \varphi_b^2 \Big). \tag{5.4}$$

In the limit $n \to 0$. Let us now proceed with the proof. The first idea consists in eliminating the non-local term of the energy, at the price of defining a new field ϕ. Using the Stratonovich identity (or completing the square for a Gaussian integral) with $\int \mathrm{d}s \int \mathrm{d}s' \delta(r(s) - r(s')) = \int dx (\delta(x - r(s)))^2$:

$$G(r,0,N,0) = \mathcal{N} \int \mathcal{D}\phi(x) \int_{r(0)=0}^{r(N)=r} \mathcal{D}r(s) exp\Big(- \frac{d}{2b^2} \int_0^N \mathrm{d}s \left|\frac{dr}{ds}\right|^2 \Big) exp\Big(- \frac{1}{2v} \int \mathrm{d}x \phi^2(x) - i \int_0^N \mathrm{d}s \phi(r(s))\Big). \tag{5.5}$$

Which can be rewritten as:

$$G(r,0,N,0) = \mathcal{N} \int \mathcal{D}\phi(x) exp\Big(- \frac{1}{2v} \int \mathrm{d}x \phi^2(x)\Big) G(r,0,N,0,\phi). \tag{5.6}$$

$G(r,0,N,0,\phi)$ is a path integral with kinetic energy + external potential, for every ϕ, and it satisfies the Bloch equation:

$$\frac{\partial G}{\partial s} = \frac{b^2}{2d} \nabla_r^2 G - i\phi(r) G; \tag{5.7}$$

Which, rearranged and with the initial conditions added, becomes:

$$\Big(\frac{\partial}{\partial s} - \frac{b^2}{2d} \nabla_r^2 + i\phi(r)\Big) G(r,0,N,0) = \delta(r)\delta(N). \tag{5.8}$$

Applying the Laplace transform with respect to N:

$$\Big(m^2 - \frac{b^2}{2d} \nabla_r^2 + i\phi(r)\Big) G(r,0,m^2,0) = \delta(r). \tag{5.9}$$

The result obtained can be compared with the formal identity:

$$G(x,x',m^2,0,\phi) = \frac{1}{Z} \int \mathcal{D}\varphi(x) \varphi(x) \varphi(x') exp\Big(- \frac{1}{2} \int \mathrm{d}x \int \mathrm{d}x' \varphi(x) G^{-1}(x,x') \varphi(x')\Big) \tag{5.10}$$

$$= \frac{1}{Z} \int \mathcal{D}\varphi(x) \varphi(x) \varphi(x') exp\Big(- \frac{1}{2} \int \mathrm{d}x \varphi(x) (m^2 - \frac{b^2}{2d} \nabla_x^2 + i\phi(x)) \varphi(x)\Big) \tag{5.11}$$

We observe that, formally in this case too, the identity holds:

$$Z^{-1} = \lim_{n\to 0} Z^{n-1} \tag{5.12}$$

In this way we can add n fields to the equation and write:

$$G(x,x',m^2,0,\phi) = \int \mathcal{D}^n\varphi(x)\varphi_1(x)\varphi_1(x')exp\Big(-\frac{1}{2}\int \mathrm{d}x \sum_{a=1}^{n}\varphi_a(x)(m^2 - \frac{b^2}{2d}\nabla_x^2 + i\phi(x))\varphi_a(x)\Big) \tag{5.13}$$

At this point we rewrite and insert the last identity in the equation of the real propagator:

$$G(x,x',m^2,0) = \int \mathcal{D}^n\varphi(x)\varphi_1(x)\varphi_1(x')exp\Big(-\frac{1}{2}\int \mathrm{d}x \sum_{a=1}^{n}\varphi_a(x)(m^2 - \frac{b^2}{2d}\nabla_x^2)\varphi_a(x)\Big) \tag{5.14}$$

$$\mathcal{N}\int \mathcal{D}\phi(x)exp\Big(-\frac{1}{2v}\int \mathrm{d}x\phi^2(x) - iv\sum_a \varphi_a\phi(x)\varphi_a\Big) \tag{5.15}$$

Completing the square in ϕ one obtains the final result:

$$G(x,x',m^2,s') = \int \mathcal{D}^n\varphi(x)\varphi_1(x')\varphi_1(x'')exp(-\beta H_{GLW}(\varphi)) \tag{5.16}$$

With

$$\beta H_{GLW}(\varphi) = \frac{1}{2}\int \mathrm{d}x\Big(\sum_{a=1}^{n}[|\boldsymbol{\nabla}\varphi_a|^2 + m^2\varphi_a^2] + v\sum_{a,b}\varphi_a^2\varphi_b^2\Big). \tag{5.17}$$

The theory obtained is therefore a theory with O(n) symmetry for $n \to 0$ and with a φ^4 perturbation. From the 4-d expansion method through the renormalisation group one can compute perturbatively the exponent ν of the general theory:

$$\nu = \frac{1}{2}(1 + \frac{1}{2}((n+2)/(n+8))\epsilon + O(\epsilon^2)) \tag{5.18}$$

Which for $n \to 0$ reduces to:

$$\nu = \frac{1}{2}(1 + \frac{1}{8}\epsilon + O(\epsilon^2)) \tag{5.19}$$

Which at first order in ϵ in d=3 comes close to the estimate given by Flory. Again through this equivalence it is possible to compute the value of γ. Going back to the SAW chain on a lattice, the stochastic process corresponding to the scaling limit is still unknown. Although the $n \to 0$ limit is difficult to formalise in this interpretation, the derivation can be made rigorous by using a fermionic field and studying the supersymmetric version of the φ^4 model.

6. Other possible computations

6.1 Exponents and scaling in D=2

An additional property, although only conjectured, in 2D is that the scaling limit of the SAW, in order to be a stochastic process, enjoys conformal invariance besides scale invariance. This allows one to apply results of CFT and of SLE. In particular one finds that, assuming conformal invariance of the model, this corresponds to $SLE_{8/3}$. Knowing that the fractal dimension for

an SLE_k is the minimum between $(2, 1 + k/8)$ one finds that for the SAW $d_f = 4/3$, and since $\nu = 1/d_f$ one deduces the result $\nu = 3/4$. From similar reasoning γ can also be obtained.

$$\nu_{2D} = 3/4, \qquad \gamma_{2D} = 43/32. \tag{6.1}$$

Note that in 2D the hexagonal lattice is the only case in which the value of $\mu = \sqrt{2 + \sqrt{2}}$ is known exactly.

6.2 Exponents and scaling in D ≥ 4

As anticipated earlier, in $D \geq 4$ the avoidance term becomes negligible with respect to the connectivity term, and the scaling limit converges to the Brownian motion typical of the classical RW. The result for $D \geq 5$ has been proved rigorously through the method of the *lace expansion*, while in $D = 4$ there are problems due to logarithmic corrections (for example of the type $log^{1/4}(N)$ for the end-to-end distance) to the power terms (analogously to what happens in Ising for the exponent α). In general one therefore finds:

$$\nu_{D\geq 4} = 1/2, \qquad \gamma_{D\geq 4} = 1. \tag{6.2}$$

6.3 Cluster expansion

As mentioned earlier, it is possible to establish an equivalence between the partition function of an ISAW and the two-point function of an O(n) ϕ^4 theory. It is possible to establish a relation between the two models also by considering the models on the discrete lattice. Considering the O(n) model on a lattice one has a partition function:

$$\sum_{\{S_i\}} exp(J \sum_{i,j} S_i \cdot S_j). \tag{6.3}$$

In the high-temperature limit $J << 1$ and it is possible to Taylor expand the exponential as sums of powers. Because of parity symmetry, the non-zero terms are characterised by the fact of having only an even number of outgoing legs, so as to form loops. In particular one can show that at first order in J a loop contributes to the sum with a factor n; in particular one has:

$$Z(L) = \sum_{L=0}^{\infty} J^L n^{\#loops}. \tag{6.4}$$

Where L is the length of the diagram. In the limit $n \to 0$ the configurations containing loops vanish and only the trivial one remains, leaving $Z = 1$. Considering instead the two-point correlation function $G_{i,j}$, the paths to be considered are those that have as endpoints S_i and S_j. One thus obtains:

$$G_{ij} = \sum_{i\to j, L} \#config.\ J^L. \tag{6.5}$$

One thus obtains the generating function for SAWs on a lattice. Estimates of this type indeed motivate defining at the lattice level quantities that turn out to be macroscopic quantities of the $O(n)$ model, which exhibit phase transitions as a function of certain parameters. For example it is possible to define the susceptibility:

$$\chi(\nu) = \int_0^{\infty} \mathrm{d}t\ C_t e^{-\nu t}. \tag{6.6}$$

So that there exists a critical value ν_c below which it diverges, and the correlation function:

$$G_\nu(x) = \int_0^\infty \mathrm{d}t \; C_t(x) e^{-\nu t}. \tag{6.7}$$

Analogous to formula (6.5).

7. References

- The mean-field approach and the Flory estimate in $D = 3$ are found in the course notes;
- "POLYMER THEORY: PATH INTEGRALS AND SCALING" by T.A. VILGIS for the equivalence between SAW and the $O(N)$ model;
- "Lectures on Self-Avoiding Walks" by Roland Bauerschmidt, Hugo Duminil-Copin, Jesse Goodman, and Gordon Slade, and "The Self-Avoiding Walk: A Brief Survey" by Gordon Slade for discussions on the case D=2, D$\geq$4 and the equivalence between SAW and WSAW.